



Introduction to Agentic Frameworks





Agenda

In this session, we'll discuss:

- What are AI Agents
 - Introduction to Agentic Frameworks
 - Creating and Using Agents using Langchain and Crew-AI
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About Me

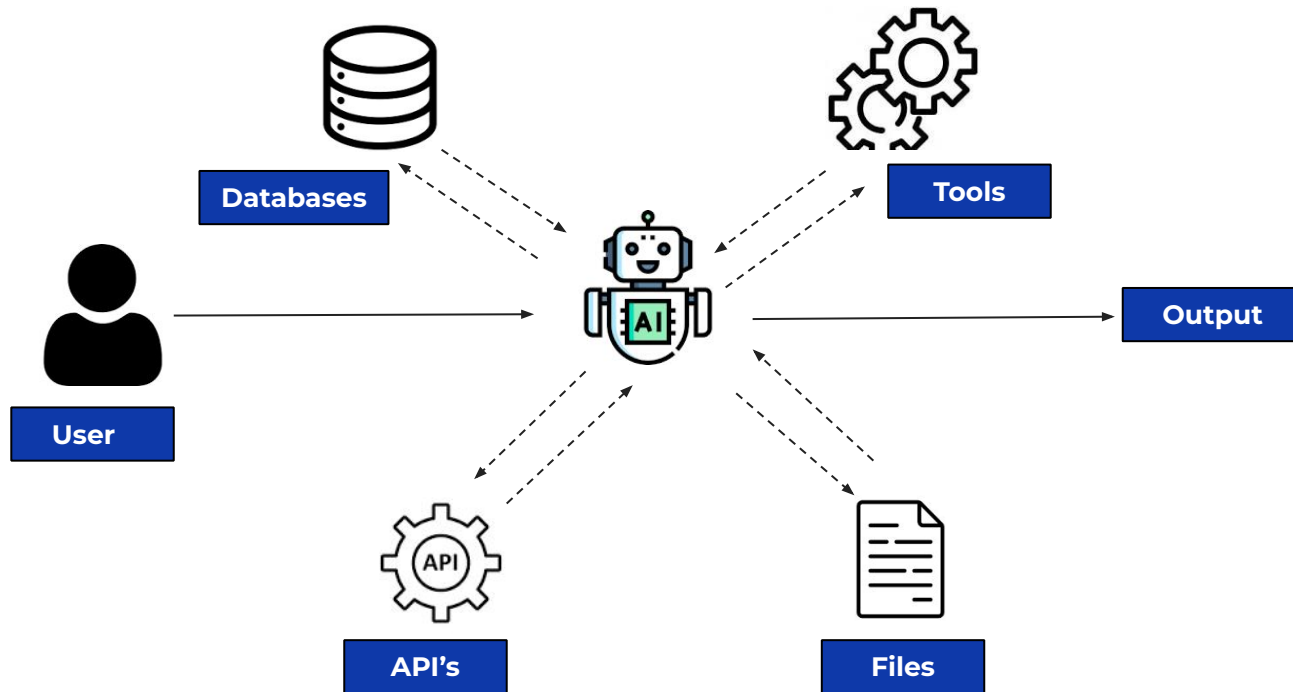
Anuj Saini is a Subject Matter Expert in the area of Natural Language Processing, Search Technologies, Statistics, Analytics, Modelling, Data Science, Data Mining and Machine Learning. He has multi-years industry experience in developing systems based on NLP, Machine Learning, and Cloud Technologies.

- **Researcher at University of Montreal, Canada**
- **Principal Data Scientist at RPX**
- **Data Science Trainer/Mentor at Great Learning**



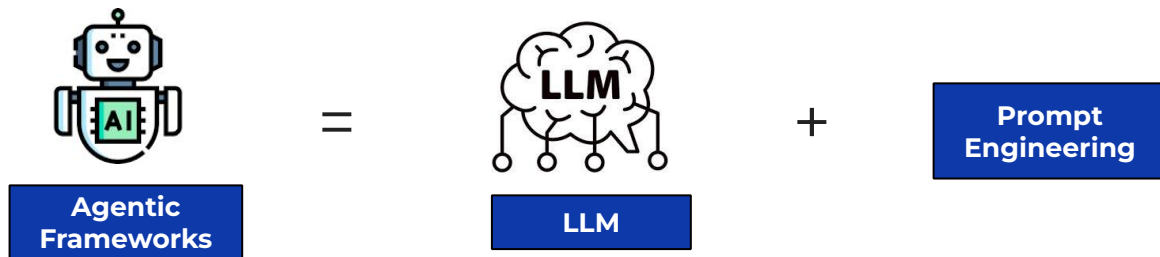
What are AI Agents?

AI Agents are autonomous or semi-autonomous systems that use artificial intelligence to perform tasks, make decisions, or solve problems without direct human intervention.

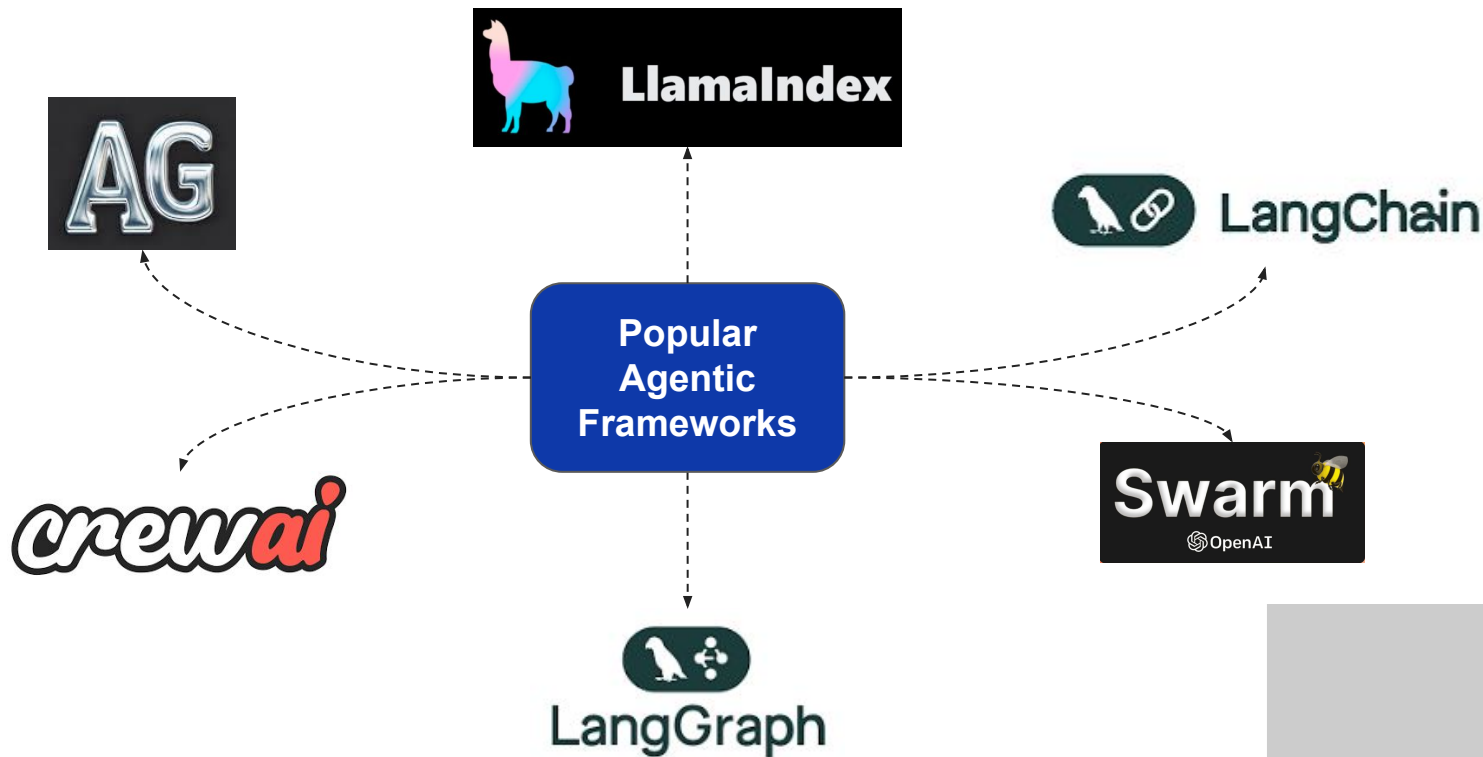


Introduction to Agentic Frameworks

- **Agentic Frameworks** serve as a conceptual **blueprint** for the design and implementation of AI agents, outlining how these autonomous systems can operate effectively to achieve specific goals.
- **Agentic Frameworks leverage Large Language Models (LLMs)** to enhance the agents' understanding of natural language, allowing them to process complex queries, generate human-like responses, and improve their overall ability to interact with users and comprehend context.



Popular Agentic Frameworks used in the Industry



Popular Agentic (Contd)

- **Langchain:** LangChain enables the creation of complex workflows by chaining together LLM calls and integrating various data sources, facilitating natural language processing tasks efficiently.
- **LangGraph:** LangGraph integrates LLMs with knowledge graphs, allowing users to visualize relationships and perform dynamic queries, enhancing information retrieval in natural language applications.
- **LlamaIndex:** LlamaIndex combines LLMs with index-based search capabilities, enabling efficient access to large datasets and providing contextually relevant responses to user queries

Popular Agentic (Contd)

- **Swarm AI:** Swarm AI utilizes decentralized algorithms based on collective behavior, enabling multiple AI agents to work together collaboratively, optimizing problem-solving and decision-making processes
- **Crew AI:** Crew AI emphasizes collaboration among AI agents, designed for applications that require teamwork and shared decision-making to achieve common goals effectively.
- **AutoGen:** AutoGen automates the generation of AI applications by providing user-friendly tools that integrate LLMs, streamlining the development process for various tasks with minimal coding.



Practical Hands-on using:

- LangChain to develop a Pandas DataFrame agent
 - Utilizing Crew AI for creating a blog writing agent
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Choosing your Learning Pathway

Program	Weeks	Who is this for?	What will you learn?	Why should you enroll?
Microsoft Generative AI for Business	16 weeks	Functional roles / professionals in a particular domain like sales, operations, finance who want to leverage Generative AI to solve their frequently occurring problems	Azure OpenAI Studio, Prompt Engineering Techniques, build workflows with Prompt Flow, solve natural language problems, RAG, Responsible AI - learning journey is primarily No Code tools based	- Gain the skills to use Generative AI tools to streamline workflows, improve decision-making, and drive business efficiency without coding expertise using Azure Platform Note: Agentic AI overview is covered as Masterclass in this program.
Generative AI for NLP	18 weeks	Advanced Course - python programming is a prerequisite - for developer roles / upskill as a data scientist - deeper conceptual depth, advanced techniques and deployment	Llama-cpp, OpenAI API, Multimodal Prompts, AI Assistant Development, Complex Agentic Workflows, Advanced RAG, Alignment Fine-tuning and Optimization, Application Deployment, LLM Ops, Cutting edge of AI	- Achieve advanced proficiency in NLP and Generative AI, equipping yourself with the expertise to solve complex problems and innovate at the forefront of Gen AI (NLP).
GenAI for Practitioner	12 weeks	Designed for tech professionals with Python skills who want to move from theory to hands-on GenAI application, build AI-powered solutions, and integrate them into existing workflows and System.	Master Generative AI by learning prompt engineering, advanced RAG, and agentic workflows using tools like LangChain, LangGraph, and CrewAI. Build and deploy real-world AI applications with automation and customization.	-Master advanced GenAI tools to build and deploy real-world AI solutions. - Learn to automate workflows and solve business problems with RAG and agentic AI.
UT-GABA	14 weeks	Designed for business and tech professionals who wish to build a strong foundation in Generative AI and acquire practical, industry-ready Generative AI skills to drive Generative AI adoption in the workplace.	Large Language Models, Prompt Engineering, RAG, LLM Fine-tuning, and Agentic AI with tools like Transformers, OpenAI, ChromaDB, and Langchain for building and deploying secure Generative AI solutions for a variety of business problems	- Build a strong foundation in Generative AI from a business and technical perspective - Design AI workflows using LLMs, RAG, and Agentic AI to extract meaningful insights and enhance business process efficiency - Evaluate risks and implement mitigation strategies to build secure Generative AI applications

Questions?



Happy Learning !

